

Assignment: Titanic Survival Prediction using Logistic Regression

Course: Machine Learning (4th Year B.Tech)

Topic: Binary Classification using Logistic Regression

Difficulty: Intermediate to Advanced

Objective

In this assignment, students will build a **Logistic Regression** model to predict whether a passenger survived the Titanic disaster using passenger information such as age, gender, passenger class, family members, and embarkation port.

Students will perform the complete Machine Learning pipeline including:

- Data Understanding
- Exploratory Data Analysis (EDA)
- Data Cleaning
- Feature Engineering
- Feature Encoding
- Feature Scaling
- Model Training
- Model Evaluation
- Hyperparameter Tuning
- Business Insights

Dataset

Link: <https://www.kaggle.com/datasets/heptapod/titanic>

Use the **Titanic Dataset** provided in the Kaggle notebook.

Target Variable:

Survived

- 0 → Passenger did not survive
- 1 → Passenger survived

Example Features:

- Pclass
- Sex
- Age
- SibSp
- Parch
- Embarked etc

Business Problem

The Titanic disaster is one of the most well-known historical datasets used to understand **binary classification**. The objective is to develop a predictive model that estimates the probability of passenger survival based on demographic and travel-related information.

Business Insights (Replace the previous section)

Assume you are a Data Scientist working for a maritime safety organization.

Answer the following:

1. Which factors most influenced passenger survival?
2. Did passenger class affect survival probability?
3. Did gender significantly impact survival?
4. What role did age play?
5. Which feature was the strongest predictor?
6. What recommendations would you provide for improving passenger safety in future maritime travel?

Assignment Tasks

Part 1: Understanding the Dataset

Perform the following:

1. Load the dataset.
2. Display first 10 rows.
3. Check dataset dimensions.
4. Display column information.
5. Check missing values.

6. Identify numerical and categorical features.
7. Display descriptive statistics.

Deliverables

- Dataset summary
- Observations (minimum 5)

Part 2: Exploratory Data Analysis

Perform:

- Class distribution
- Correlation heatmap
- Histograms
- Boxplots
- Pairplots (optional)
- Outlier analysis

Write observations:

- Which feature is highly correlated?
- Is the dataset balanced?
- Which variables seem useful?

Part 3: Data Preprocessing

Perform

- Missing value handling
- Label Encoding / One-Hot Encoding
- Feature Scaling
- Train-Test Split (80-20)

Explain why each preprocessing step is necessary.

Part 4: Logistic Regression Model

Train a Logistic Regression model.

Use Scikit-Learn.

Display

- Model coefficients
- Intercept
- Feature importance (based on coefficients)

Part 5: Model Evaluation (20 Marks)

Evaluate using

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- Classification Report

Also explain:

- Which metric is more important for this dataset?
- Why?

Report Format

The report should contain

1. Introduction
2. Problem Statement
3. Dataset Description
4. EDA
5. Data Preprocessing
6. Model Building
7. Evaluation
8. Hyperparameter Tuning
9. Business Insights
10. Conclusion

Maximum length: **8–10 pages**

Submission Guidelines

Submit:

- Jupyter Notebook (.ipynb)
- PDF Report
- README.md
- Dataset (if permitted)
- Images used in the report